

Advanced Mastery Worksheet: Mech. Osc.

Focus: Energy Graphics, Damping & Linearization

Exercise 1: Dynamic Equation & Kinematic Deduction

A horizontal elastic pendulum consists of a solid block of mass m attached to a massless ideal spring of stiffness k . The horizontal surface is perfectly frictionless. The block is displaced from its equilibrium position ($x = 0$) and released.

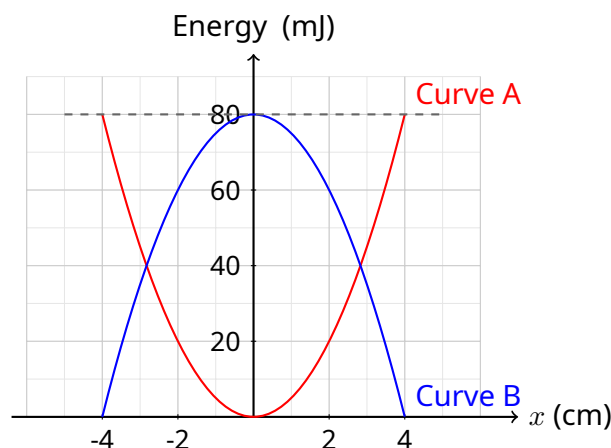
1.1 By applying the principle of conservation of mechanical energy, establish the second-order differential equation governing the motion of the block.

1.2 Verify that $x(t) = X_m \cos(\omega_0 t + \varphi)$ is a valid solution to this differential equation, and deduce the expression for the proper angular frequency ω_0 .

1.3 For a specific system where $k = 40 \text{ N/m}$ and $m = 0.1 \text{ kg}$, the block is pulled to an initial abscissa $x_0 = 2.0 \text{ cm}$ and pushed towards the equilibrium with an initial velocity $v_0 = -0.4 \text{ m/s}$ at $t = 0$. Calculate ω_0 , the amplitude X_m , and the initial phase φ .

Exercise 2: Multi-Method Analysis from Energy Graphs

The graph below represents the variations of two energy components of an undamped horizontal oscillator as a function of the position x (in cm).



2.1 Identify which curve (A or B) corresponds to the Kinetic Energy (KE) and which corresponds to the Elastic Potential Energy (EPE). Justify your answer physically.

2.2 Extract directly from the graph the total Mechanical Energy (ME) and the amplitude of motion (X_m).

2.3 Demonstrate mathematically that when the speed of the block is $v = V_{max}/\sqrt{2}$, the kinetic and potential energies are strictly equal. Deduce the exact values of the position x where this occurs for the graphed system.

Exercise 3: The Pseudo-Periodic Regime

The same horizontal oscillator is now placed on a rough surface where it experiences a viscous friction force $\vec{f} = -h\vec{v}$ (where h is a positive constant and $\vec{v}$ is the velocity vector).

3.1 Apply Newton's Second Law to establish the new differential equation governing the damped motion of the block.

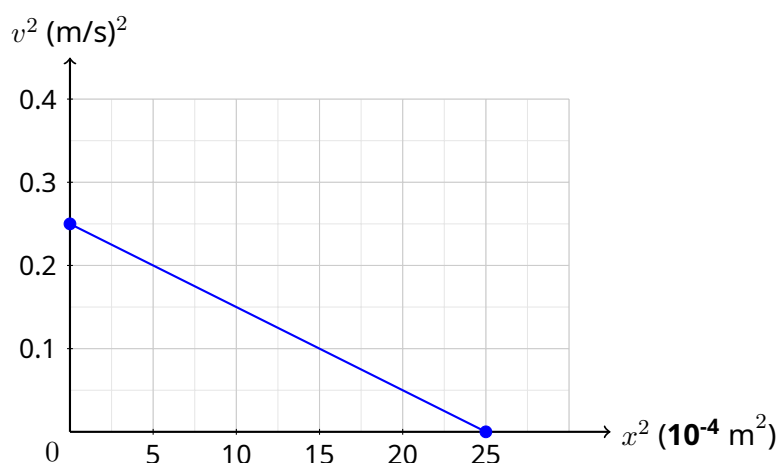
3.2 Knowing that $ME = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$, show that the rate of change of the mechanical energy with respect to time is $\frac{d(ME)}{dt} = -hv^2$. What does the negative sign signify physically?

Exercise 4: Linearization of the Phase Space

Returning to the isolated (undamped) system, the conservation of energy ensures that $\frac{1}{2}mv^2 + \frac{1}{2}kx^2 = ME_{max}$.

4.1 Manipulate this energy equation to express v^2 as a linear function of x^2 in the form $y = mx + c$. Identify the theoretical expressions for the slope m and the y-intercept c in terms of ω_0 and X_m .

4.2 A student sets up an undamped oscillator and plots the squared velocity v^2 as a function of the squared position x^2 . The resulting graph is the straight line shown below.



- Using the coordinates of the intercepts, calculate the mathematical slope of the line.
- Deduce the proper angular frequency ω_0 of this specific oscillator.
- Deduce the amplitude X_m (in cm) given to the system by the student.